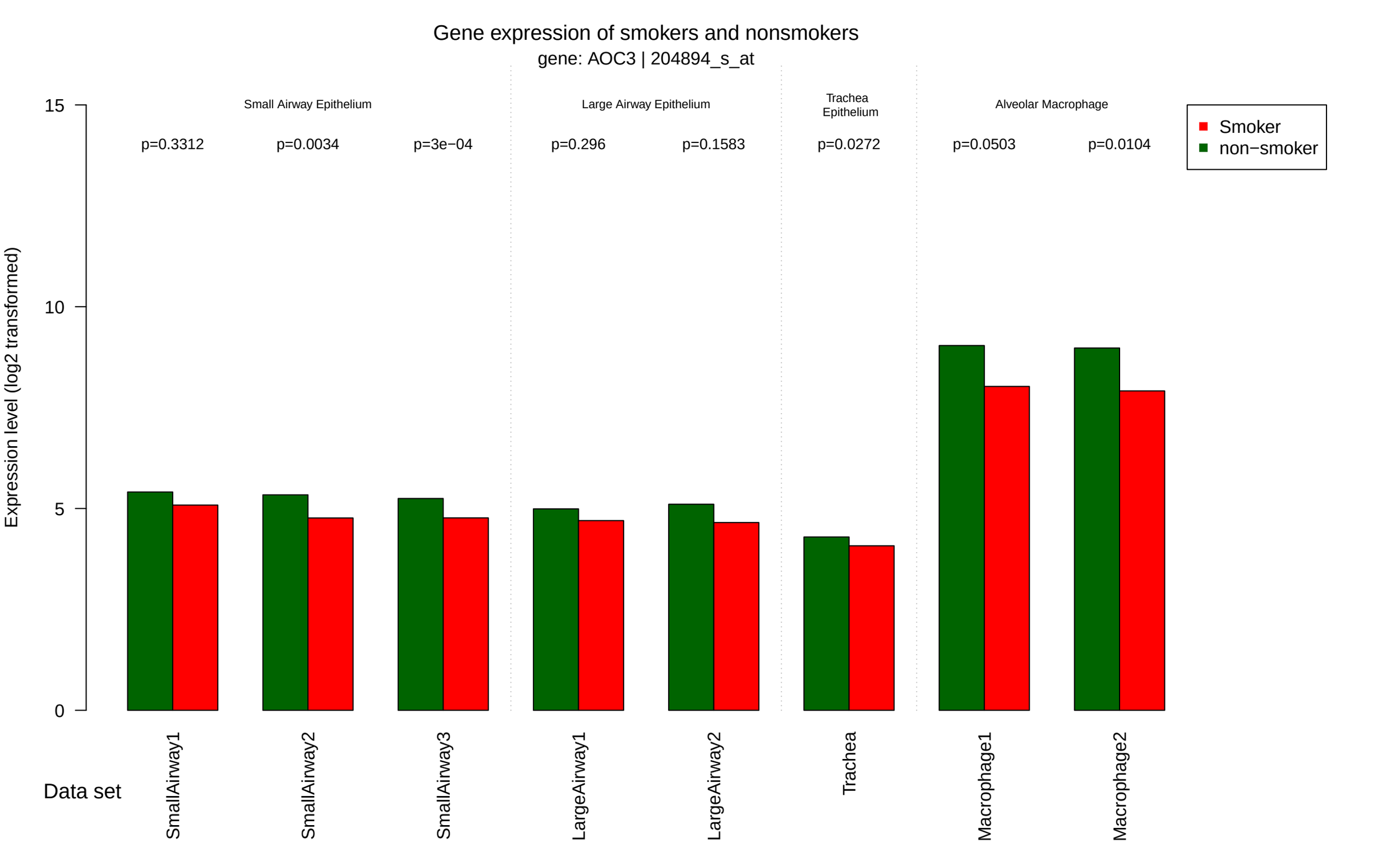
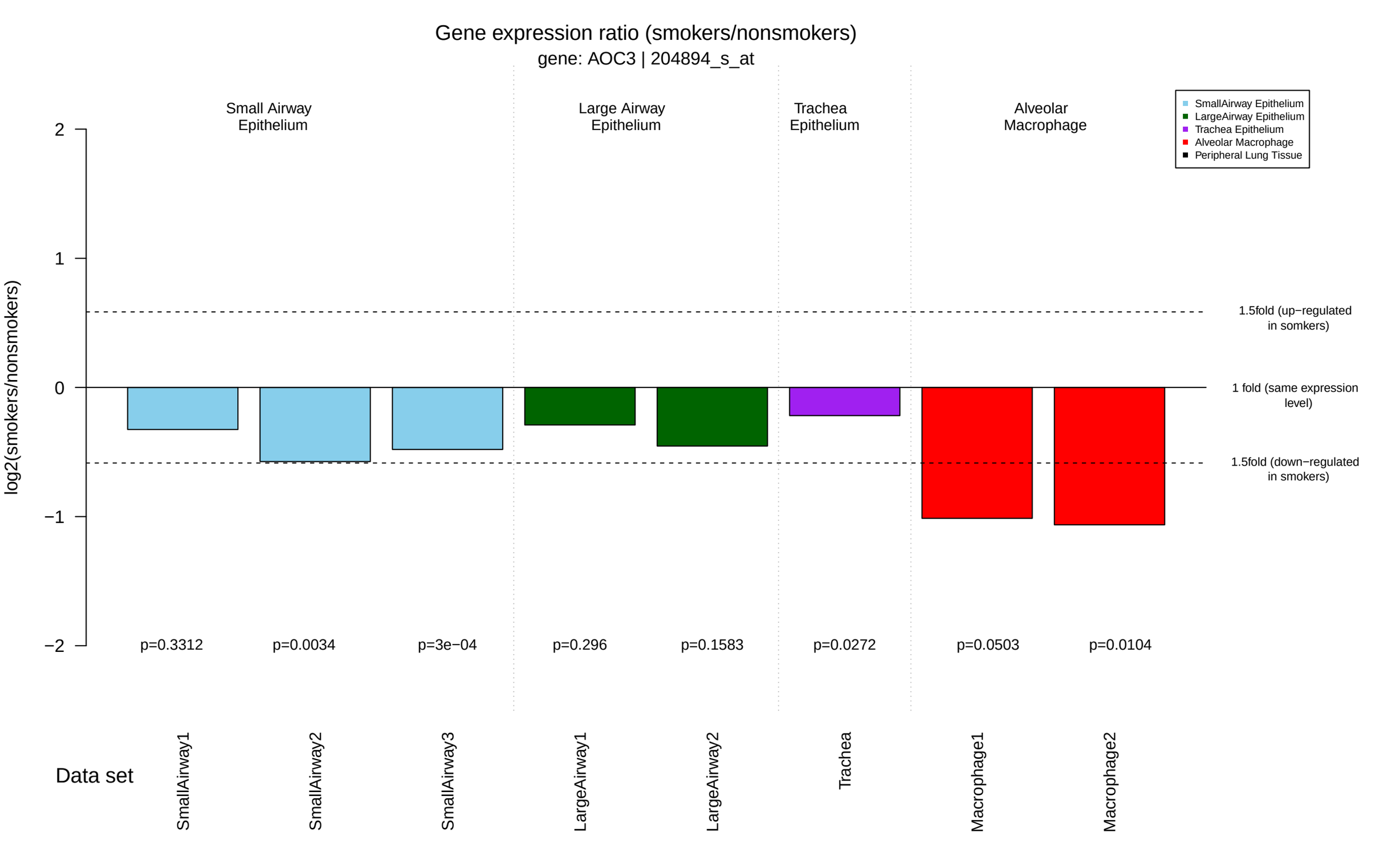
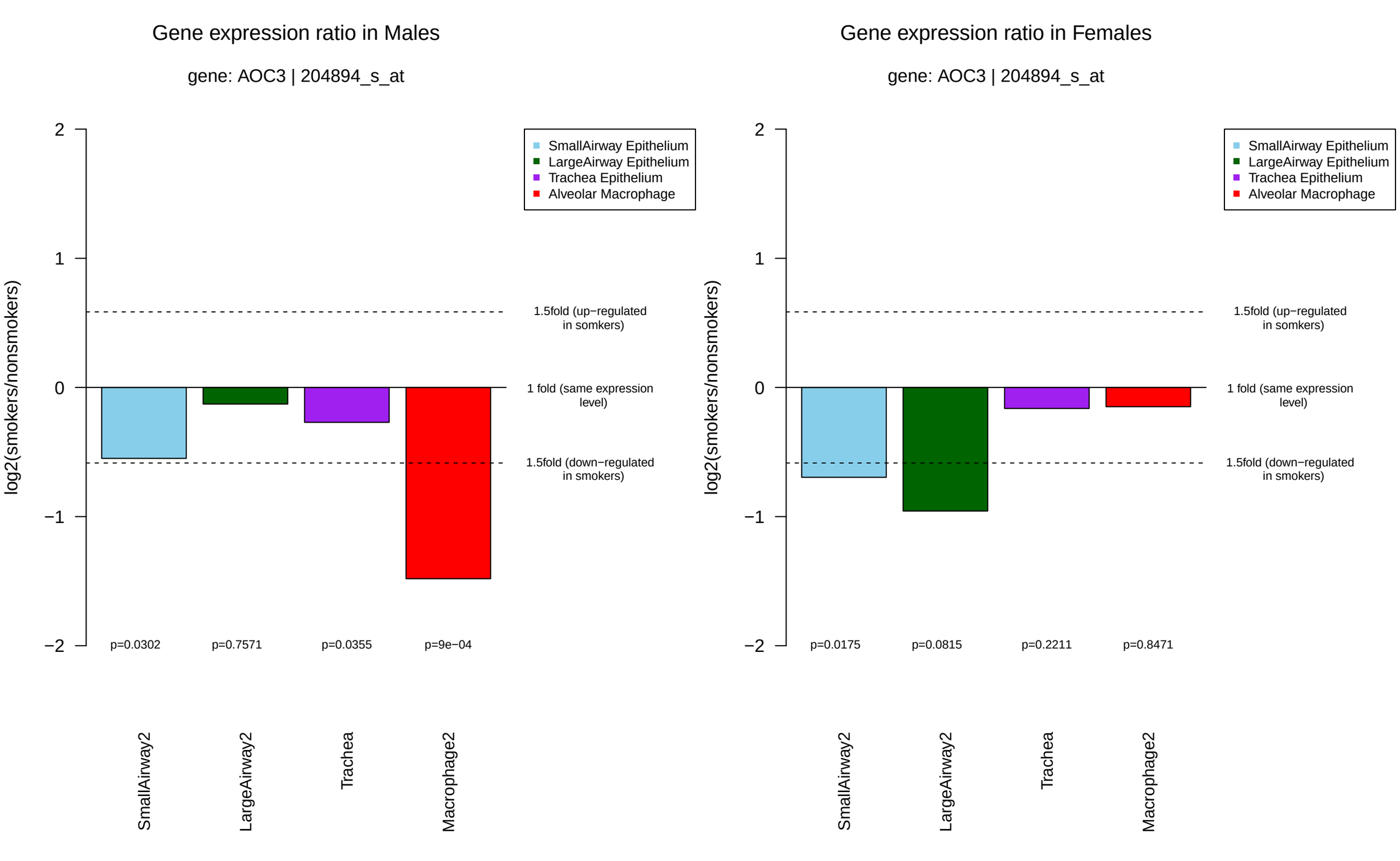
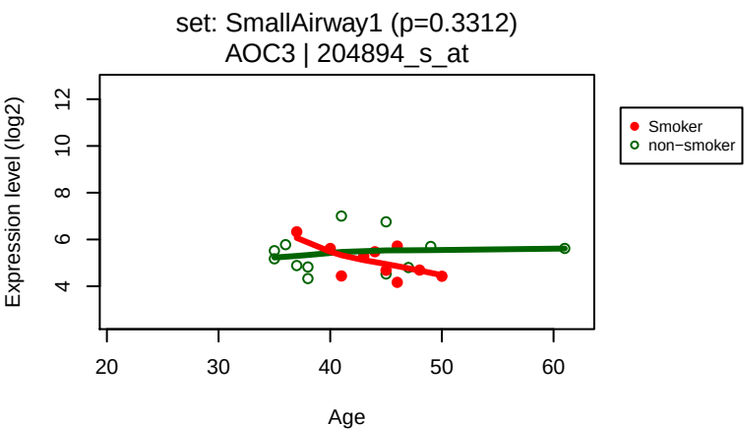
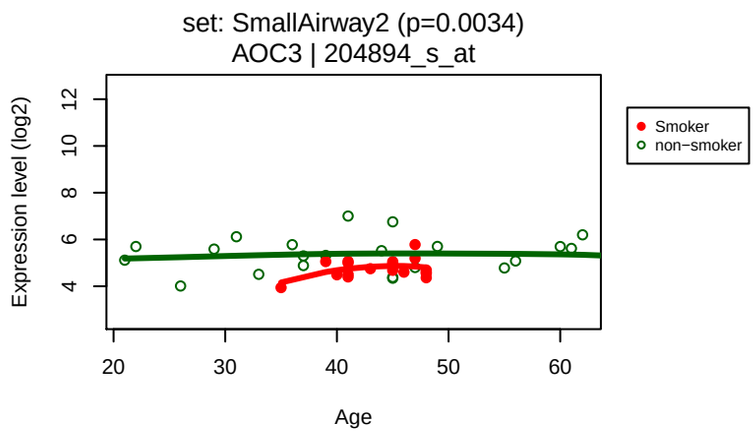
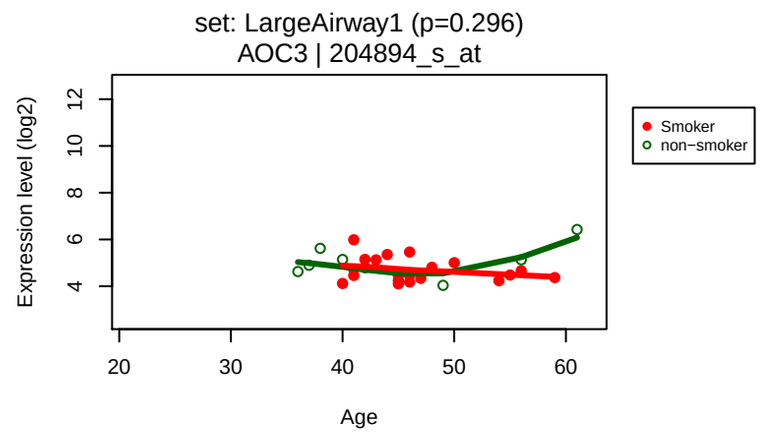

Gene expression of smokers and nonsmokers
gene: AOC3 | 204894_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3312
p=0.0034
p=3e−04
p=0.296
p=0.1583
p=0.0272
p=0.0503
p=0.0104
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

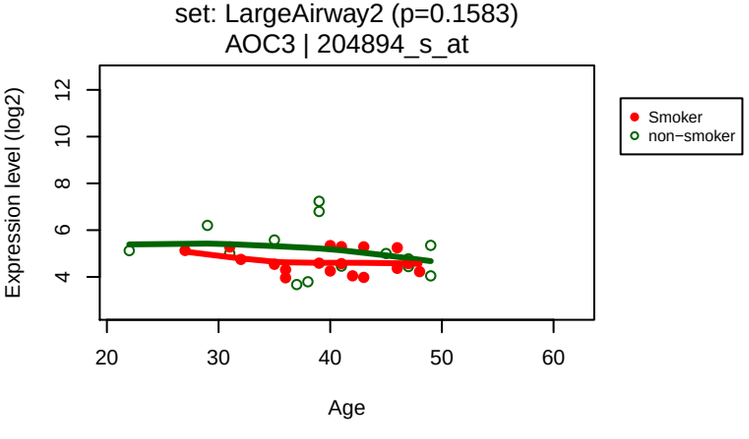

Gene expression ratio (smokers/nonsmokers)
gene: AOC3 | 204894_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3312
p=0.0034
p=3e−04
p=0.296
p=0.1583
p=0.0272
p=0.0503
p=0.0104
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

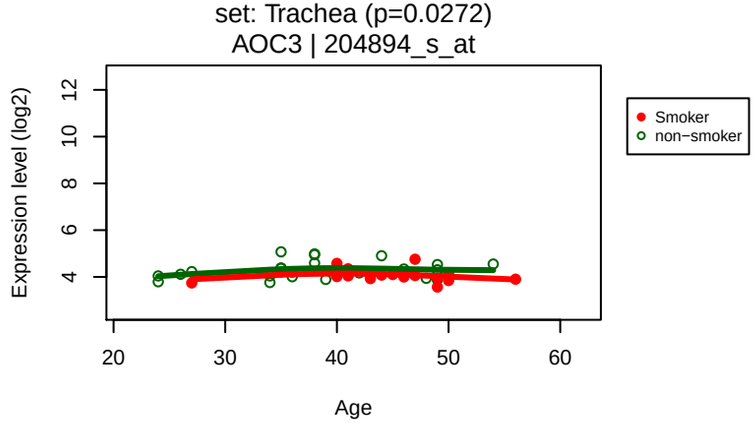

Gene expression ratio in Males
gene: AOC3 | 204894_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0302
p=0.7571
p=0.0355
p=9e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AOC3 | 204894_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0175
p=0.0815
p=0.2211
p=0.8471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

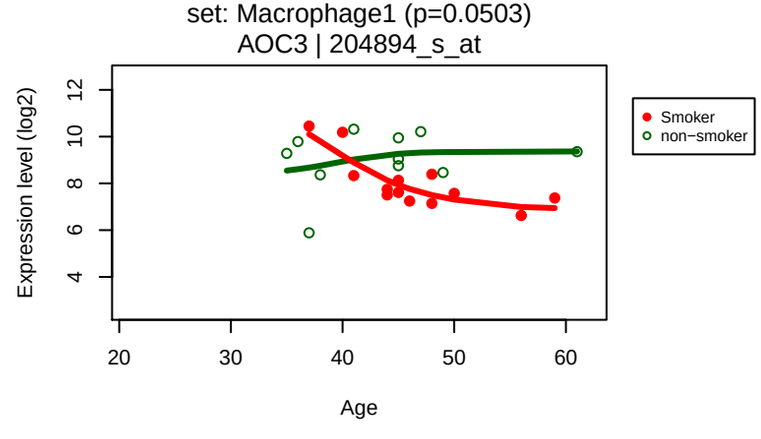

set: SmallAirway1 (p=0.3312)
AOC3 | 204894_s_at
Expression level (log2)
Age
Smoker
non−smoker

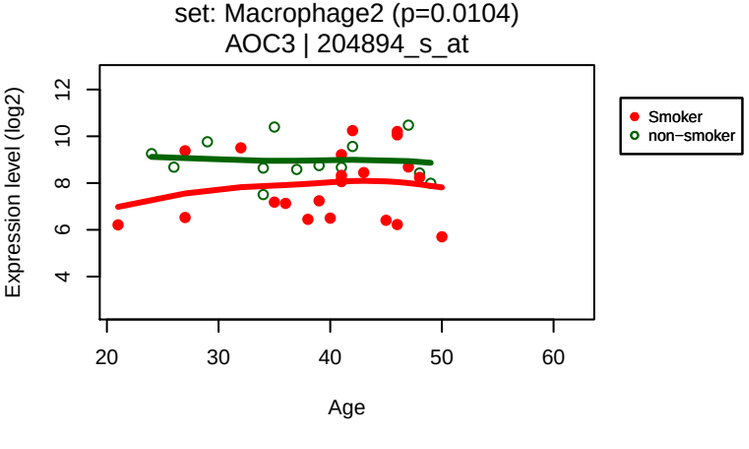

set: SmallAirway2 (p=0.0034)
AOC3 | 204894_s_at
Expression level (log2)
Age
Smoker
non−smoker

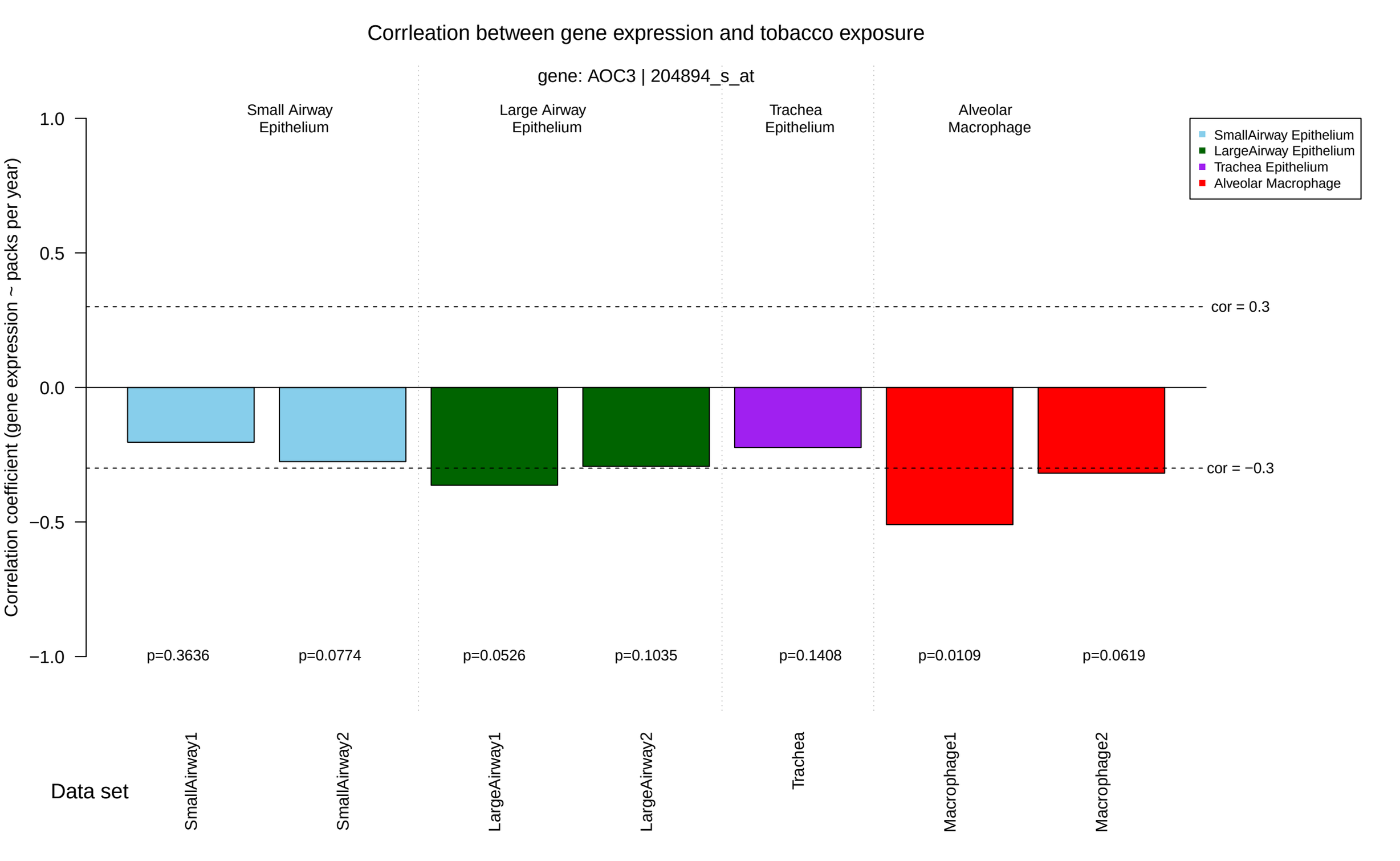

set: LargeAirway1 (p=0.296)
AOC3 | 204894_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1583)
AOC3 | 204894_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0272)
AOC3 | 204894_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0503)
AOC3 | 204894_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0104)
AOC3 | 204894_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AOC3 | 204894_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3636
p=0.0774
p=0.0526
p=0.1035
p=0.1408
p=0.0109
p=0.0619
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2